Predicting Medical Insurance Charges Using Linear Regression

Course: Machine learning technques
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1. Introduction

Health insurance companies rely on statistical models to estimate medical costs based on demographic and lifestyle characteristics. Accurate prediction of insurance charges helps companies:

- Set fair premiums
- Identify high-risk clients
- Manage financial risk
- Improve pricing strategies

This report develops and evaluates both **Simple and Multiple Linear Regression models** using the *Medical Cost Personal Dataset* to predict medical insurance charges.

The dependent variable is:

- **Charges** (medical insurance cost)

Independent variables include:

- Age
- Sex
- BMI
- Children
- Smoker
- Region

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2. Methodology

The analysis followed these steps:

1. Data Acquisition and Understanding
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis (EDA)
4. Linear Regression Modeling
5. Model Evaluation
6. Business Interpretation and Recommendations

All analysis was conducted in Python using Jupyter Notebook.

PART A: Data Acquisition and Understanding

Dataset Overview

The dataset represents medical insurance costs for individuals based on personal and lifestyle characteristics.

- Observations: 1338
- Variables: 7

Dependent Variable

- charges

Independent Variables

- age
- sex
- bmi
- children
- smoker
- region

Data Types

- Numerical: age, bmi, children, charges

- Categorical: sex, smoker, region

The dataset contains no structural issues.

PART B: Data Cleaning and Preprocessing

Missing Values

- Checked using .isnull().sum()
- Result: No missing values found

Therefore, no imputation was required.

Duplicate Records

- Checked using .duplicated().sum()
- A small number of duplicates were found .
- Duplicates were removed to prevent bias in model training.

Outliers

BMI and Charges

Boxplots revealed:

- BMI contains mild extreme values (> 45)
- Charges contain strong right-tail outliers (values > 50,000)

These were retained because:

- High medical charges are realistic (e.g., smokers with complications)
- Removing them would bias the model and reduce real-world relevance

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Encoding Categorical Variables

Categorical variables were converted using **Dummy Encoding**:

- Smoker = smoker_yes (1 = Yes, 0 = No)
- sex = sex_male

- region → region dummies

Dummy encoding prevents incorrect ordinal interpretation.

Feature Scaling

Scaling is **NOT mandatory** for Linear Regression because:

- OLS regression is scale-invariant for prediction
- Coefficients remain interpretable in original units

However, scaling would be useful if using regularized regression.

PART C: Exploratory Data Analysis (EDA)

Summary Statistics

Key findings:

- Mean Age: =39 years
- Mean BMI: 30
- Mean Charges: $13,270
- Max Charges: $63,770

Charges show high variance.

Distribution of Charges

Histogram shows:

- Strong right skewness
- Most people pay below $20,000
- Small group pays extremely high costs

Conclusion: Charges are positively skewed.

Relationship Analysis

Age against Charges (Scatter Plot)

- Positive linear trend
- Older individuals generally pay more

BMI against Charges

- Weak positive relationship

- High BMI slightly increases charges

according to Smoker Status (Boxplot)

Major finding:

- Smokers pay dramatically higher charges
- Median charges for smokers are more than double non-smokers

Smoking appears to be the strongest predictor.

Correlation Matrix

Strongest correlations with charges:

- Smoker status (strong positive)
- Age (moderate positive)
- BMI (weak-moderate)

Smoker has the highest correlation.

PART D: Linear Regression Modeling

Section 1: Simple Linear Regression

Independent variable chosen: **Age**

Model Equation:

Charges=β0+β1(Age)Charges = $\beta_0 + \beta_1$(Age)Charges=β0+β1(Age)

Estimated equation:

Charges=−11938+257(Age)Charges = -11938 + 257(Age)Charges=−11938+257(Age)

Interpretation

Intercept (-11938):

- Predicted charge when age = 0
- Not practically meaningful but mathematically necessary

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Slope (257):

For each 1-year increase in age, medical charges increase by approximately $257.

$R^2 \approx 0.09$

- Age alone explains about 9% of the variation in charges.
- Indicates weak explanatory power alone.

Section 2: Multiple Linear Regression

Model including all predictors.

Full Regression Equation (Simplified Form)

Charges=β0+β1(Age)+β2(BMI)+β3(Children)+β4(Smoker)+...Charges $= \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{BMI}) + \beta_3(\text{Children}) + \beta_4(\text{Smoker}) + \ldots$Charges=β0+β1(Age)+β2(BMI)+β3(Children)+β4(Smoker)+…

Key Results

- $R^2 \approx 0.75$
- Adjusted $R^2 \approx 0.74$

Model explains about 75% of variation in charges.

Significant Variables ($p < 0.05$)

- Age
- BMI
- Smoker
- Children

Smoking has the largest coefficient.

Smoking Effect

Smokers pay approximately **$23,000 more** on average than non-smokers, holding other factors constant.

This is the strongest impact variable.

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PART E: Model Evaluation and Assumptions

Train-Test Split

- 70% Training

  30% Testing

Performance on Test Data

- RMSE ≈ 6000
- MAE ≈ 4200
- $R^2 \approx 0.73$

Model performs well and generalizes effectively.

Assumptions

Linearity

Scatter plots and residuals show approximate linear relationships.

Satisfied.

Homoscedasticity

Residual plot shows slight variance increase at higher predictions.

Mild heteroscedasticity observed.

Normality of Residuals

Histogram approximately normal but slightly right-skewed.

Acceptable for large samples.

Multicollinearity (VIF) All VIF < 5.

No multicollinearity problem.

PART F: Interpretation and Business Recommendations

Which factor increases charges the most?

- Smoking status has the largest effect.

If BMI increases by 1 unit?

Charges increase by approximately $339 (based on coefficient).

Recommendations

Recommendation 1:

Introduce higher premiums for smokers due to significantly higher medical costs.

Recommendation 2:

Implement wellness programs targeting BMI reduction to lower long-term insurance payouts.

Recommendation 3:

Offer preventive care incentives for older clients to reduce high age-related costs.

These recommendations are directly supported by statistically significant predictors.

Conclusion

This study successfully developed and evaluated Linear Regression models to predict medical insurance charges.

Key findings:

- Smoking is the strongest determinant of charges.
- Age and BMI also significantly increase costs.
- The multiple regression model explains approximately 75% of variation in charges.
- The model performs well on unseen data.

The results provide strong evidence that insurance pricing should heavily account for smoking status and health indicators.

References

- Kutner, M. H., Nachtsheim, C. J., Neter, J. (2004). Applied Linear Statistical Models.
- Kaggle: Medical Cost Personal Dataset.
- Python Documentation (pandas, sklearn, statsmodels)